

Customer Retention & Churn Analysis

Project Overview

This project focuses on analyzing customer churn and retention patterns for a subscription-based business. The main objective is to identify the factors responsible for customer loss and provide actionable recommendations to improve customer retention.

Objectives

- Analyze customer churn trends
 - Identify high-risk customer groups
 - Study customer retention patterns
 - Calculate Customer Lifetime Value (CLV)
 - Generate business insights and recommendations
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Tools & Technologies Used

- Python
 - Pandas
 - Matplotlib
 - Jupyter Notebook
 - Excel / Power BI (Optional)
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Dataset Information

The dataset contains customer subscription details such as:

- Customer ID
 - Gender
 - Contract Type
 - Monthly Charges
 - Payment Method
 - Tenure
 - Churn Status
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Key Analysis Performed

1. Churn Rate Analysis

Calculated the percentage of customers who left the subscription service.

2. Contract Type Analysis

Compared churn rates among:

- Month-to-Month contracts
- One-Year contracts
- Two-Year contracts

3. Tenure Analysis

Studied how customer duration affects retention.

4. Payment Method Analysis

Identified payment methods associated with higher churn.

5. Customer Lifetime Value (CLV)

Estimated long-term revenue generated from customers.

Key Insights

- Month-to-month customers have the highest churn rate.
 - Customers with higher monthly charges are more likely to churn.
 - New customers are at higher risk during the initial months.
 - Long-term contracts improve customer retention significantly.
 - Electronic check payment users show higher churn behavior.
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Business Recommendations

1. Offer discounts for long-term subscriptions.
 2. Improve onboarding experience for new customers.
 3. Introduce loyalty reward programs.
 4. Enhance customer support services.
 5. Use predictive analytics to identify churn risks early.
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Project Deliverables

- Customer Churn Analysis Report (PDF)
 - Jupyter Notebook / Python Code
 - Dataset File
 - Dashboard / Charts
 - README.md
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Repository Structure

FUTURE_DS_02/

| — README.md

| — FUTURE_DS_02_Task2_Report.pdf

| — customer_churn_analysis.ipynb

| — customer_churn_dataset.csv

| — dashboard.png

Conclusion

This project successfully identified important churn factors and retention drivers in a subscription-based business. The analysis can help organizations improve customer satisfaction, reduce churn, and increase long-term revenue growth.

Author

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